

Lecture 4: Measures of Dispersion

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Learning Outcomes

- Explain the meaning of dispersion through examples
- Define various measures of dispersion
- Calculate various measures of dispersion of raw and grouped data
- Illustrate the properties of variance and standard deviation.
- Define and compute the coefficient of variation.

Introduction

Some definitions of dispersion are given below

- ◇ **Brooks & Dick:** Dispersion or spread is the degree of the scatter or variation of the variable about a central value.
- ◇ **Spiegel:** The degree to which numerical data tend to spread about an average value.
- ◇ **Simpson & Kajka:** The measurement of the scatterness of the mass of figures in a series about an average is called measure of variation or dispersion

Dispersion (also known as scatter, spread or variation) measures the extent to which the items vary from some central value.

Range

- ◇ $Range = Max - Min$
- ◇ **Example:** A sample of five executives received the following bonuses last year (\$000) : 14.0, 15.0, 17.0, 16.0, 15.0
- ◇ $Range = 17 - 14 = 3$
- **Example:** The weight example (weight.csv)

Properties of Range

Properties

- Only two values are used in its calculation.
- It is influenced by an extreme value (non-robust).
- It is easy to compute and understand.

Limitations of Range

- It is based only on two items and does not cover all the items in a distribution
- It is influenced by an extreme value (non-robust).
- It fails to give any idea about the pattern of distribution

Quartile Deviation

- The Median is a number which divides the data set into two equal parts
- The Quartile is a number which divide the data set into quarters
- Quartile Deviation (QD) = $IQR/2$
- Interquartile Range (IQR) = $Q_3 - Q_1$

where Q_1 is first quartile and Q_3 is third quartile

Four-step process to find QD

Step 1. Sort the data in a nondecreasing order

Step 2.

- ◇ The number on the $n/4$ position is the first quartile Q_1
- ◇ The number on the $(3n)/4$ position is the third quartile Q_3

Step 3. $IQR = Q_3 - Q_1$

Step 4. $QD = IQR/2$

Example

- ◇ **Example:** A sample of five executives received the following bonuses last year (\$000) : 14.0, 15.0, 17.0, 16.0, 15.0
- Arranging the data in ascending order gives:
14.0, 15.0, 15.0, 16.0, 17.0
- Q_1 position is $n/4 = 5/4 = 1.25 \simeq 1^{st} position = 14.0$.
- Q_3 position is $\frac{3n}{4} = \frac{3 \times 5}{4} = 3.75 \simeq 4^{th} position = 16.0$.
- $IQR = Q_3 - Q_1 = 16 - 14 = 2$
- $QD = IQR/2 = 2/2 = 1$

Properties of Quartile Deviation QD

Merits

- In certain respects it is superior to the range as a measure of dispersion.
- Can be used in open ended distributions.
- Useful in erratic or badly skewed data

Limitations

- It fails to cover all the items in a distribution.
- It is not amenable to mathematical manipulation.
- It varies widely from sample to sample based on the same population.
- Since it is a positional average, it is not considered as a measure of dispersion

Mean Deviation (*MAD*)

- This is the average of the absolute values of deviations from the mean
- The Mean Absolute Deviation *MAD* of a set of n numbers

$$MAD = \frac{|x_1 - \mu| + \dots + |x_n - \mu|}{n}$$

$$MAD = \frac{\sum_i^n |x_i - \mu|}{n}$$

Example

- ◇ **Example:** A sample of five executives received the following bonuses last year (\$000) : 14.0, 15.0, 17.0, 16.0, 15.0
- ◇ **Problem:** Determine the MAD.
- ◇ **Solution:**

$$\bar{x} = \frac{14.0 + 15.0 + 17.0 + 16.0 + 15.0}{5} = 15.4$$

$$MAD = \frac{|14.0 - 15.4| + \dots + |15.0 - 15.4|}{5} = 1.22$$

Properties of MAD

Merits

- All values are used in the calculation.
- It is not unduly influenced by large or small values (robust)
- The absolute values are difficult to manipulate.

Limitations

- It is not capable of further algebraic treatment.
- At times it may fail to give accurate results.
- Ignoring of the signs makes the method non algebraic.

Standard Deviation (SD)

- This is the average of deviation from the mean
- The Standard Deviation is the square root of variance
- The variance σ^2 of a set of N numbers as population:

$$\sigma^2 = \frac{(x_1 - \mu)^2 + \dots + (x_n - \mu)^2}{N}$$

$$\sigma^2 = \frac{\sum_i^N (x_i - \mu)^2}{N} = \frac{\sum_i^N x_i^2}{N} - \mu^2 \rightsquigarrow \text{proof}$$

Sample Standard Deviation (s)

- The variance of a set of n numbers as sample:

$$s^2 = \frac{(x_1 - \bar{x})^2 + \dots + (x_n - \bar{x})^2}{n - 1}$$

$$s^2 = \frac{\sum_i^n (x_i - \bar{x})^2}{n - 1} = \frac{\sum_i^n x_i^2 - n\bar{x}^2}{n - 1} \rightsquigarrow \text{proof}$$

Example

- ◇ **Example:** A sample of five executives received the following bonuses last year (\$000) : 14.0, 15.0, 17.0, 16.0, 15.0
- ◇ **Problem:** Determine the standard deviation.
- ◇ **Solution:**

$$\bar{x} = \frac{14.0 + 15.0 + 17.0 + 16.0 + 15.0}{5 - 1} = 15.4$$

$$s^2 = \frac{(14.0 - 15.4)^2 + \dots + (15.0 - 15.4)^2}{5 - 1} = 1.3$$

$$s = \sqrt{1.3} = 1.14$$

Properties of SD

Merits

- All values are used in the calculation.
- It is not extremely influenced by outliers

Limitations

- It is difficult to compute
- It gives more weight to extreme items and less to those which are near the mean.

Range for grouped data

- The range of a sample of data organized in a frequency distribution is computed by the following formula:
$$\text{Range} =$$

upper limit of the last class – lower limit of the first class

Standard Deviation for Grouped Data

- The Standard Deviation is the square root of variance
- The variance of data organized in a frequency distribution is computed by the following formula:

$$\sigma^2 = \frac{\sum_i^N f_i(x_i - \mu)^2}{N} = \frac{\sum_i^N f_i x_i^2}{N} - \mu^2 \rightsquigarrow \text{proof}$$

- where f_i is the class frequency and x_i is the class midpoint for Class $i = 1 \dots N$.

Example

① **Example:** Recall the weight example from Chapter 2:

Weight	F	mid point (x_i)	$f_i x_i$	$f_i x_i^2$
130 - 136	2	133	266	35378
137 - 143	5	140	700	98000
144 - 150	9	147	1323	194481
151 - 157	11	154	1694	260876
158 - 164	6	161	966	155526
165 - 171	5	168	840	141120
172 - 178	2	175	350	61250
	40		6139	946631

- The mean for the grouped data is:

$$\bar{x} = \frac{6139}{40} = 153.475$$

- The Variance s^2

$$s^2 = \frac{\sum_i^n f_i(x_i - \bar{x})^2}{n - 1} = \frac{\sum_i^n f_i x_i^2 - n\bar{x}^2}{n - 1} \rightsquigarrow \text{proof}$$

$$s^2 = \frac{946631 - 40 \times 153.475^2}{40 - 1} = 114.056$$

$$s = \sqrt{114.056} = 10.67$$

THANK YOU

QUESTIONS